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humans exactly. What matters more is whether it can understand the situation, consider the available information, and choose an action that gives the best expected result.

Humans can also make mistakes, so simply copying human thinking or behavior is not always the best goal. The rational-agent approach is more practical because it focuses on making good decisions and achieving goals effectively.

Chapter # 02

Intelligent Agents

1 Suppose that the performance measure is concerned with just the first T time steps of the environment and ignores everything thereafter. Show that a rational agent's action may depend not just on the state of the environment but also on the time step it has reached.

A rational agent chooses the action that gives the best expected performance. If performance is measured only for the first T time steps, then the amount of time remaining can affect which action is best.

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For example, suppose an agent must reach a destination before time T. from the same location:

- If enough time remains, it may choose a safe but longer route.
- If only one or two time steps remain, it may need to choose a shorter route to reach the goal before the deadline.

Therfore, even when the state of the environment is the same, the rational action may be different at different times. Hence, a rational decision can depend on both the current state and the current time step.

2.2 Let us examine the rationality of various vacuum-cleaner agent functions.

a) Show that simple vacuum-cleaner agent function described in Fig 2.3 is indeed rational under the assumptions listed on page 38.

The simple vacuum agent follow these rules:

- If the current square is dirty → Suck
- If square A is clean → move Right
- If square B is clean → move Left

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Under the given assumptions, this behavior is rational. Sucking immediately cleans a dirty square, while moving from a clean square allows the agent to check and clean the other square. Since movement has no cost and clean squares remain clean, these actions maximize the number of clean squares during the agent's lifetime.

b) Describe a rational agent function for the case in which each movement costs one point. Does the corresponding agent program require internal state?

If movement has a cost, continuously moving between already-clean squares would waste points. A better agent would:

- Suck if the current square is dirty.
- Visit the other square if its condition is not yet known or it needs cleaning.
- Stop when it knows that both squares are clean.

Yes, internal state is required because the agent must remember whether the other square has already been visited and cleaned.

c) Discuss possible agent designs for the cases in which clean squares can become dirty and the geography of the environment is unknown. Does it make sense for the agent to learn from its experience? If so what should it learn?

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b) Yes, learning would be useful in both cases.

If clean squares can become dirty again, the agent should learn where and how often dirt usually appears so that it can decide when to check those areas again.

If the geography is unknown, the agent should explore the environment and build an internal map, learning the locations of different squares, boundaries and obstacles.

By learning from experience, the agent can improve its future decisions and operate more efficiently.

2-3 For each of the following assertions, say whether it is true or false and support your answer with examples or counterexamples where appropriate.

a) An agent that senses only partial information about the state cannot be perfectly rational.

False. An agent can still be rational with partial information if it chooses the best action using the information available to it. For example if the same action is best in all possible hidden states, the agent can choose it rational

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a) There exist task environments in which no pure reflex agent can behave rationally.

True. A pure reflex agent uses only the current percept. In some environments, the correct action also depends on previous information. Such an agent needs memory or internal state.

b) There exists a task environment in which every agent is rational.

True. If every possible action gives exactly the same performance, then no action is better than another. Therefore every agent behaves rationally.

c) The input to an agent program is same as the input to the agent function.

False. The agent program receives the current ~~perc~~ percept, whereas agent function maps the complete percept sequence to an action.

d) Every agent function is implementable by some program/machine combination.

False. Some possible agent functions may require unlimited memory or computation and therefore cannot be implemented by a real finite machine.

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f) Suppose an agent selects its action uniformly at random from the set of possible actions. There exists a deterministic task environment in which this agent is rational.

True. If available actions gives the same result or performance, choosing randomly among them is rational.

g) It is possible for a given agent to be perfectly rational in two distinct task environments.

True. The same agent can be rational in different environments if its actions are optimal in both.

h) Every agent is rational in an unobserval environment.

False. Even without sensors, some actions may still be better than others because of the agent's prior knowledge.

i) A perfectly rational poker-playing agent never loses.

False. Poker contains uncertainty and hidden information. A rational agent chooses actions that maximize its expected success, but it can still lose a particular game.

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4 For each of the following activities, give a PEAS description of the task environment and characterize it in terms of the properties listed in Section 2.3.2.

Playing Soccer:-

Performance: Win the match, score goals, prevent opponent goals, follow rules.
Environment: Field, ball, teammates, opponents, referee, goalposts.
Actuators: Running, kicking, passing, jumping, communicating
Sensors: Vision, hearing, position, movement sensing

Exploring the Subsurface Oceans of Titan:-

P: Collect useful scientific data, explore safely, conserve energy.
E: Ocean, Ice, rocks, current, underwater terrain.
A: Thursters, robotic arms, sampling equipment, communication devices.
S: Cameras, sonar, temperature, pressure and chemical sensors.

- Properties: Partially observable, single-agent, stochastic, sequential, dynamic, continuous, largely unknown.

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- **Shopping for Used AI Books on Interne**

P: Find required books in a good condition a low price and reliable delivery.
E: Websites, seller, buyers, book listing, payment sy
A: Search, click, compare, message seller, bid or purchase.
S: Search results, prices, descriptions, rating and availability.

- Properties: Partially observable, multiagent, stochasti sequential, dynamic, discreate, partially unknow

- **Playing a Tennis Match :-**

P: Win points and ultimately win the match while following the rules.
E: Court, ball, opponent, net, weather conditions.
A: Running, serving, hitting, positioning.
S: Vision, hearing and body-position sensing.

Properties: Partially observable, multiagent, stochastic sequential, dynamic, continuous, known.

- **Practicing Tennis Against a Wall :-**

P: Maintain successful returns and improve accura and control.
E: Wall, court, racket and ball
A: Moving & hitting the ball.
S: Vision & body position sensing.

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- Properties: Fully observable, single-agent, mostly deterministic, sequential, dynamic, continuous, known.

Performing a High Jump:-

P: Clear the greatest possible height without violating rules.
E: Runway, bar, landing area, weather conditions.
A: Running, Jumping, body movements.
S: Vision, balance and body-position sensing.

Properties: Partially observable, single-agent, stochastic, sequential, dynamic, continuous, known.

Knitting a Sweater:-

P: Produce the correct sweater: with minimum errors material waste.
E: Yarn, needles, and partially completed sweater.
A: Moving needles, pulling & arranging yarns.
S: Vision and touch sensors.

Properties: Fully observable, single-agent, deterministic, sequential, static, mostly discrete, known.

Bidding on an Item at an Auction:-

P: Obtain the desired item at the best acceptable price.
E: Auction system, item, seller, and other bidders.
A: Place, increase, or stop bids.

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S: Current price, other bids, time remaining and auction information.

Properties: Partially observable, multiagent, stochastic sequential, dynamic, discrete, known.

2.5 Define in your own words the following terms: agent, agent function, agent program, rationality, autonomy, reflex agent, model-based agent, goal-based agent, utility-based agent, learning agent.

1 **Agent**: An agent is anything that perceives its environment through sensors and acts on that environment through actuators.

2. **Agent-Function**: An agent function specifies which action an agent should take for every possible percept sequence.

3. **Agent - Program**: An agent-program is the actual program or code that implements the agent function on a machine.

4. **Rationality**: Rationality means choosing the action that is expected to give the best performance, based on the available informa and knowledge.

5. **Autonomy**: It is the ability of an agent to operate and improve using its own experien

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rather than depending completely on knowledge given by its designer.

5. **Reflex Agent:** An reflex agent selects an action directly from the current percept using condition - action rules.

7. **Model-Based Agent:** A model-based agent maintains an internal state or model of the environment, allowing it to remember information that may not be visible in the current percept.

8. **Goal-Based Agent:** A goal-based agent chooses actions by considering which actions will help it reach a desired goal.

1. **Utility-Based Agent:** A utility-based agent compares possible outcomes and selects the one that provides the highest expected usefulness or benefit.

0. **Learning Agent:** A learning agent can improve its behavior from experience instead of depending only on fixed knowledge given at the start.

5 This exercise explores the differences between agent functions and agent programs.

) Can there be more than one agent program

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implements a givent agent function? Give an example, or show why one is not possible.

Ans Yes. Different programs can implement the same agent function as long as they produce the same action for every possible percept sequence. For example, two vacuum-agent programs may be written using different code structures but still follow the same rule: if dirty suck; otherwise move to the other square.

b) Are there agent functions that cannot be implemented by any agent program?

Ans Yes. Some agent functions may require unlimited memory or computation, which cannot be provided by a real finite computer.

c) Given a fixed machine architecture, does each agent program implement exactly one agent function?

Ans Yes. On a fixed architecture, a particular agent program produces a specific action for each possible percept sequence and therefore implements one agent function.

d) Given an architecture with n bits of storage, how many different possible agent programs are there?

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Each bit can have two values, 0 or 1. Therefore with n bits: 2^n different bit patterns, and hence at most 2^n different programs can be stored.

) Suppose we keep the agent program fixed but speed ~~on~~ up the machine by a factor of two. Does that change the agent function?

Normally, No. The same percept sequence will still produce the same action; the program will simply produce the result faster.

Write pseudocode agent programs for the goal-based and the utility-based agents.

Goal-Based Agent :-

```
Function Goal-Based-Agent (percept)
    state ← UPDATE-STATE (state, percept)

    if goal is not defined
        goal ← FORMULATE-GOAL (state)

    actions ← possible actions from state
    choose an action that leads toward the goal.

    return action
```

A goal-based agent consider future results & chooses an action that helps it reach its required goal.

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b) Utility-Based Agent :-

```
function UTILITY-BASED-AGENT (percept)
    state ← UPDATE-STATE (state, percept)

    actions ← possible actions from state

    for each action
        calculate expected utility of its result

    return action with highest expected utility.
```

* A utility-based agent not only checks whether a goal can be achieved, but also compares different possible outcomes and chooses the one with the highest utility.

2-8 Implement a performance-measuring environment simulator for the vacuum-cleaner world depicted in Fig 2.2 & specified on page 38. Your implementation should be modular so that the sensors, actuators and environment characteristics (size, shape, dirt, placement, etc) can be changed easily. (Note: for some choices of programming languages and OS there are already implementations in the online code repository.)

A simple modular simulator can be written as :

```
Environment :
    location = A or B
```

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status[A] = Clean or Dirty
status[B] = Clean or Dirty

function PERCEPT()
    return [location, status[location]]

function EXECUTE(action)
    if action = Suck
        status[location] = Clean

    else if action = Right and location = A
        location = B

    else if action = Left and location = B
        location = A

function UPDATE-SCORE()
    if A is clean
        score = score + 1

    if B is clean
        scor = score + 1

repeat for 1000 time steps:
    percept ← PERCEPT()
    action ← AGENT(percept)
    EXECUTE(action)
    UPDATE-SCORE()

The simulator is modular because the environment, sensors, actions and performance calculation are kept separate, so they can be modified easily.

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2-9 Implement a simple reflex agent for the vac environment in Ex 2.8. Run the environment wit this agent for all possible initial dirt configurat and agent locations. Record the performance sco for each configuration and the overall averag score.

Simple Reflex Agent :-

```
function REFLEX-VACUUM-AGENT(location, status)
    if status = Dirty
        return suck

    else if location = A
        return Right
    else
        return Left
```

Using the simulator from Ex 2.8, the possible initi configurations are:

Initial Location	A	B	Sco
A	Clean	Clean	200
B	Clean	Clean	200
A	Dirty	Clean	200
B	Dirty	Clean	199
A	Clean	Dirty	199
B	Clean	Dirty	200
A	Dirty	Dirty	199
B	Dirty	Dirty	199

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Therefore:

Average Score = $\frac{15994}{8}$ = $\boxed{1999.25}$

Thus, the simple reflex agent performs very well in this simple deterministic two-square vacuum environment.

Q Consider a modified version of the vacuum environment in Ex 2.8, in which the agent is penalized one point for each movement?

1) Can a simple reflex agent be perfectly rational for this environment? Explain.

Ans No. A simple reflex agent only knows the current percept. When its current square is clean, it cannot know whether the other square is dirty or already clean.

Therefore, it may move unnecessarily and lose one point even when both squares are already clean.

2) What about a reflex agent with state? Design such an agent.

Ans Yes. A reflex agent with state can remember the condition of previously visited squares.

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if current square is Dirty
Suck

else if other square is unknown or Dirty
Move to other square

else if both squares are known to be clean
No OPeration

* Because it remembers the state of both squares, it can stop moving once everything is clean and avoid unnecessary movement penalties.

c) How do your answers to a) and b) change if agent's perspective give it the clean/dirty status of every square in the environment?

b) In this case a simple agent can be perfectly rational because every percept provides the complete condition of the environment.

It can:

- Suck if the current square is dirty.
- Move if another square is dirty.
- Do nothing if every square is clean.

Therfore, internal state is no longer necessary because the required info is already available in the current percept.

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